



Q1. What is the difference between RJ45 and RJ11 connectors? Suresh

A1. RJ stands for registered jack, which is related to cable connectors. It describes a standardised telecommunication network interface for connecting voice and data equipment. Two of the most common jacks are RJ11 and RJ45, each having its own specific purpose. Main differences between the two are their actual use and number of wires.

RJ11 is used in telephone sets, while RJ45 is used in networking, where computers or other network elements are connected to each other.

RJ11 has two or four wires, whereas RJ45 has eight. This makes RJ11 smaller than RJ45.



Fig. 1: Typical RJ11 and RJ45 connectors

Q2. Why are power supply pins in an operational amplifier (op-amp) denoted as Vcc and Vee, instead of +V and -V? Vinaya Kumar K.

A2. Almost all integrated circuits (ICs) have at least two pins, known as power supply pins, which connect to power supply lines of the circuit. Op-amps

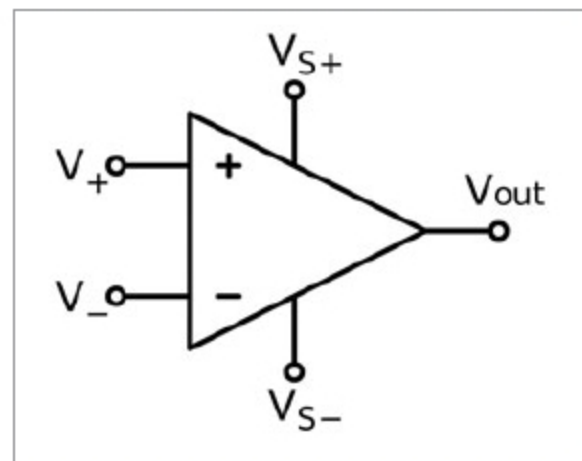


Fig. 2: Op-amp circuit symbol (Credit: www.wikipedia.org)

are widely-used electronic components in a vast array of consumer, industrial and scientific devices. The circuit diagram symbol for an op-amp with power, input and output pins is shown in Fig. 2.

The most common op-amp is bipolar-transistor-based op-amp 741

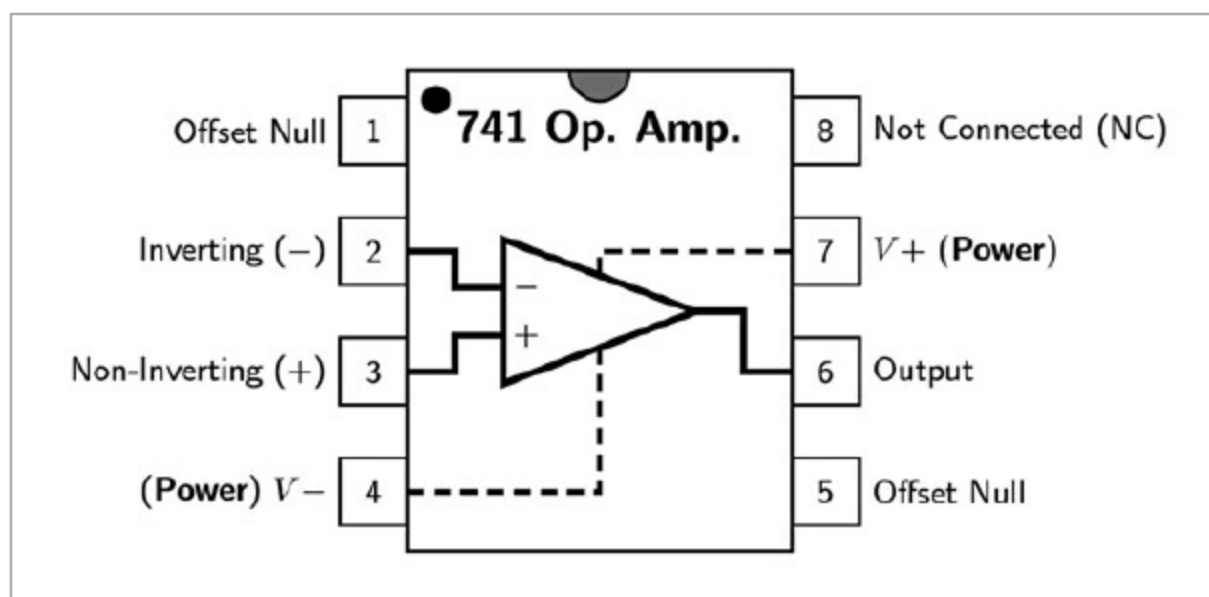


Fig. 3: Popular op-amp 741 IC pin diagram (Credit: www.wikipedia.org)

IC. The generic pin diagram of 741 IC is shown in Fig. 3.

Labelling of IC pins varies depending on the IC family and manufacturer. So, there is no fixed standard representation for power supply pins. Some manufacturers use positive power supply pins as Vcc or +Vs or Vs+, and negative power supply pins as Vee or -Vs or Vs-.

Here, Vcc stands for voltage at the common collector, which denotes voltage supplied to collector of the transistor.

Vee stands for voltage at the common emitter, which denotes voltage at emitter of the transistor.

Whereas, +V or V+ denote non-inverting input, and -V or V- denote inverting input of the op-amp.